TITANIC DATA ANALYSIS PROJECT

Internship Task: Data Analysis with Python

1. Introduction

The Titanic Data Analysis project uses Python to analyze passenger information from the Titanic disaster. The main aim is to understand passenger survival patterns based on gender and passenger class.

2. Objective

- To clean the Titanic dataset.
- To handle missing values.
- To calculate survival rates by gender.
- To compare survival rates across passenger classes.
- To visualize the results using bar charts.

3. Tools and Technologies

- Python
- Google Colab
- Pandas
- Matplotlib
- Titanic dataset

4. Data Cleaning

Missing values in the Age, Embarked, and Fare columns were filled using appropriate statistical methods. The median was used for numerical columns, while the most common value was used for the Embarked column.

5. Data Analysis

The dataset was analyzed to compare survival rates by gender and passenger class. Grouping and statistical calculations were performed using Pandas.

6. Data Visualization

Bar charts were created to visualize survival rates by gender and passenger class. These charts make it easier to compare survival patterns.

7. Key Findings

- Female passengers had a higher survival rate than male passengers.
- First-class passengers generally had a higher survival rate than second- and third-class passengers.
- Data cleaning helped prepare the dataset for analysis.

8. Conclusion

This project helped me learn how to clean data, handle missing values, calculate survival rates, and create charts using Python. The analysis showed that survival patterns differed by gender and passenger class.

9. Learning Outcomes

I gained practical experience with Python, Pandas, data analysis, and data visualization using Google Colab.

```
Missing values:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64

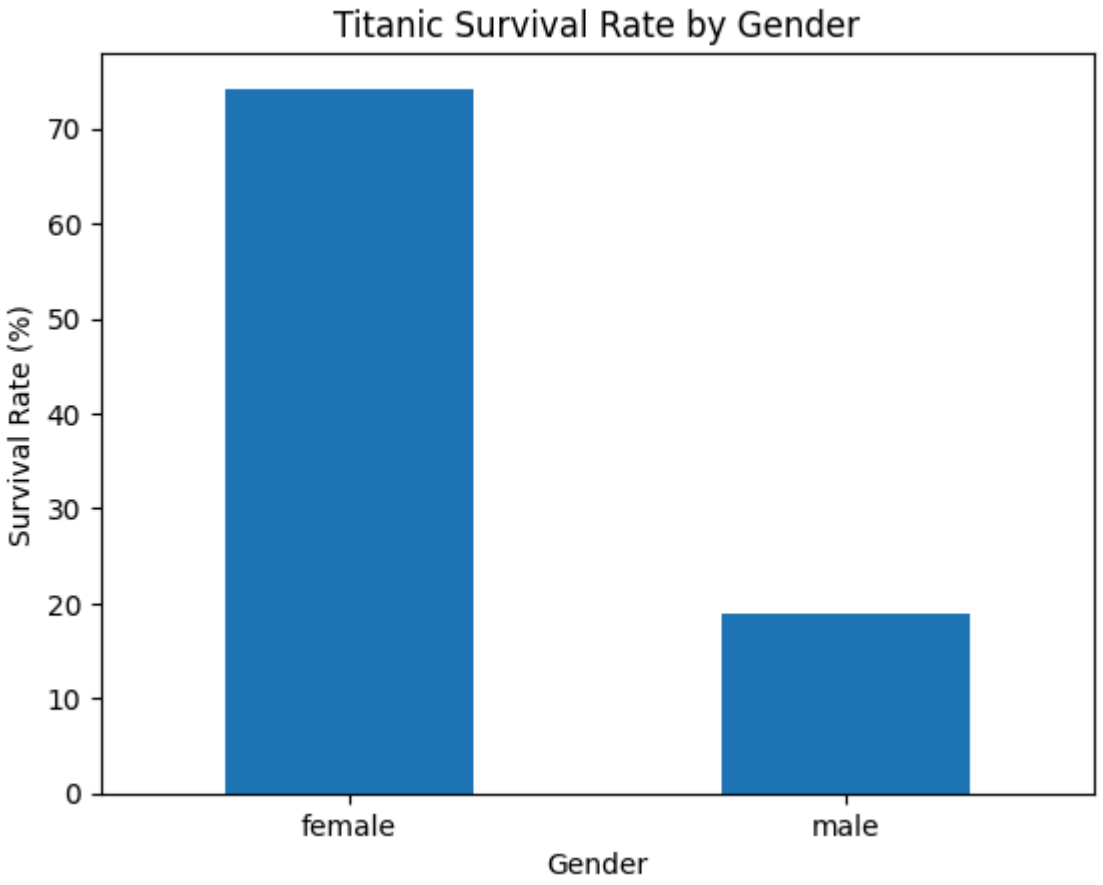
Duplicate rows: 0
```

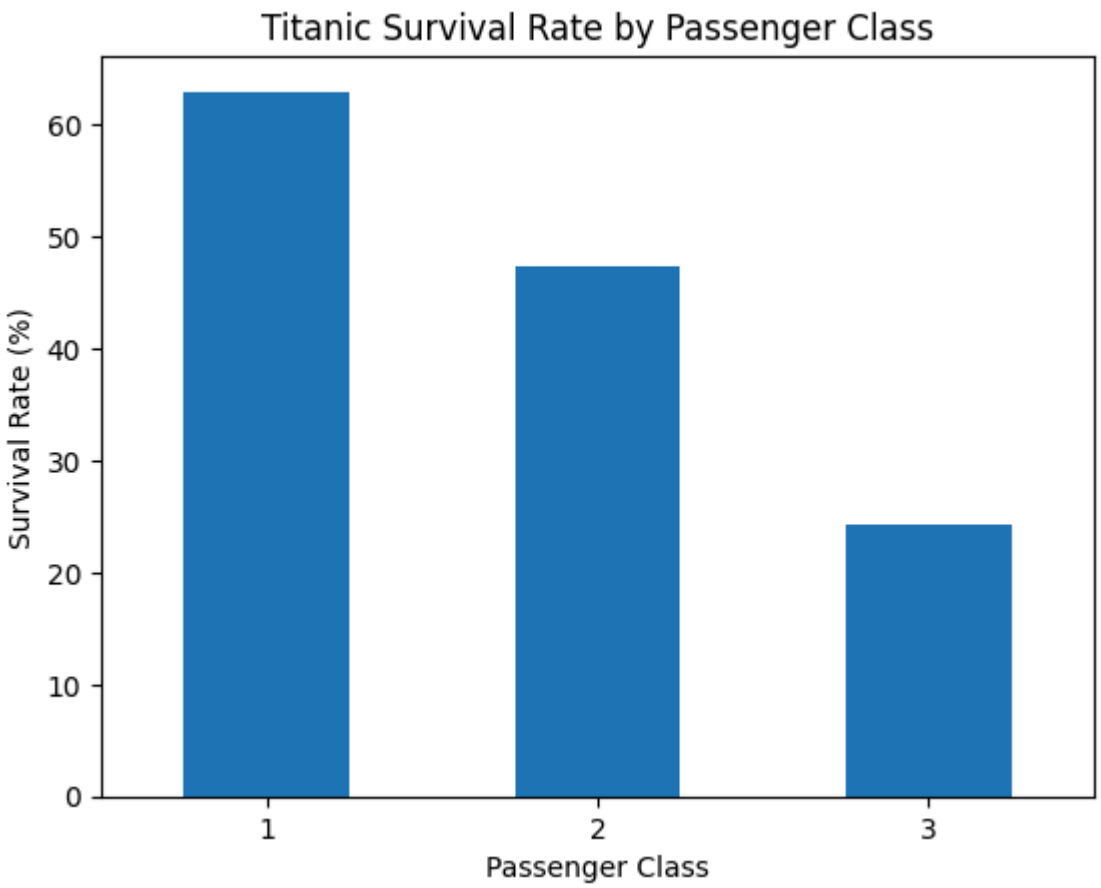
```
Survival count by gender:
Sex
female    233
male      109
Name: Survived, dtype: int64

Survival rate by gender (%):
Sex
female    74.203822
male      18.890815
Name: Survived, dtype: float64
```

```
Survivors by passenger class:
Pclass
1    136
2     87
3    119
Name: Survived, dtype: int64

Survival rate by passenger class (%):
Pclass
1    62.962963
2    47.282609
3    24.236253
Name: Survived, dtype: float64
```




Titanic Survival Rate by Passenger Class
Survival Rate (%)
0
10
20
30
40
50
60
1
2
3
Passenger Class

```
••• ===== TITANIC DATA ANALYSIS SUMMARY =====

    Total passengers: 891

    Total survivors: 342

    Overall survival rate (%): 38.38

    Survival rate by gender (%):
    Sex
    female    74.20
    male      18.89
    Name: Survived, dtype: float64

    Survival rate by passenger class (%):
    Pclass
    1    62.96
    2    47.28
    3    24.24
    Name: Survived, dtype: float64

    Missing values in Age: 0
    Missing values in Embarked: 0
    Missing values in Fare: 0
```